Dear Editor,

Please find enclosed our manuscript, *Tipping the Climate Equilibrium: Tensor-Based Game Theory for Identifying Critical Coalitions in Climate Policy Negotiations*, submitted for consideration in Environmental and Resource Economics.

The paper develops a tensor-based extension of multiplayer game theory to identify small coalitions of non-G20 nations capable of shifting the cooperative equilibrium in international climate negotiations. The central contribution is a formal connection between spectral bifurcation conditions for a third-order influence tensor and a tractable coalition scoring rule, with an exact analytical result (Proposition 1) establishing the Z-eigenvalue sensitivity of the proxy tensor to coalition activation. Empirically, the framework is applied to a real-data pipeline covering 175 non-G20 countries across 263 multilateral environmental agreements (1950–2017), augmented with composite vulnerability scores from the ND-GAIN Country Index and WRI Aqueduct 4.0. A REINFORCE policy gradient agent using Gumbel top-K sampling discovers coalitions that outperform exhaustive random search by 14.7%, identifying Uruguay, Tunisia, Greece, and Algeria as consistently high-scoring coalition anchors.

This paper is well suited to Environmental and Resource Economics. The work engages directly with the game-theoretic literature on international environmental agreements and coalition stability (Barrett, 1994; Carraro and Siniscalco, 1993; Hovi et al., 2015), extends that literature with a formal tensor-based framework, and applies it to a policy-relevant empirical setting. The vulnerability-weighted reward function connects the formal modelling to climate equity considerations of current interest in the literature.

This manuscript is original work and is not under review at any other journal. No conflicts of interest exist. AI-assisted writing tools were used during manuscript preparation; this is disclosed in the acknowledgements. The paper is submitted double-anonymously in compliance with the journal policy. GitHub repository links have been withheld from the manuscript and will be provided upon acceptance. All data and code are ready for public release in accordance with the journal data policy.

Yours sincerely,

Paul Schleifer
Independent Researcher, London, United Kingdom
ORCID: 0009-0004-7972-3566
paul$_{s}chleifer@hotmail.com$